

# Roll for the Galaxy: Mancala Dice Variant

This is the primary Roll prototype in this repository. It replaces the Dice Cup, roll, and assignment steps with a five-section mancala loop. The archived phase-battery rules remain in `docs/roll_phase_battery_rules.md`.

Use the physical dice from the base box as mancala stones. Die faces do not matter. Die color is still identity for gained dice, lost dice, color bonuses, and end-game scoring.

## Design Thesis

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Dice stay in the mancala economy. They do not become workers, construction progress, or Goods.

The loop has five sections:

```
Explore -> Develop -> Settle -> Produce -> Ship -> Explore
```

Core economy:

- Sowing moves stones and selects a phase.
- Stones in a phase section set that phase's strength.
- Credits pay for Develop and Settle.
- Good markers track produced Goods on Worlds.
- Spent is off-board and holds overflow, lost dice, and newly gained dice that cannot fit in their matching section.

This keeps the table state readable: the mancala board shows phase pressure, Credits show construction power, and Good markers show shipping inventory.

## Components

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Each player needs:

- one five-section mancala board;
- one off-board Spent area;
- Credits;
- VP chips;
- Good markers;
- their normal tableau, Development queue, and World queue.

Solo mode also needs:

- five Dummy Phase cards: Explore, Develop, Settle, Produce, Ship;
- one Dummy Goods marker;
- a shared VP pool.

## Dice Colors

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|        |                            |
|--------|----------------------------|
| Blue   | Explore identity           |
| Brown  | Develop identity           |
| Red    | Military / Settle identity |
| Green  | Produce identity           |
| Purple | Ship identity              |

|        |                |
|--------|----------------|
| White  | Wild identity  |
| Yellow | Alien identity |

White and Yellow have no native entry section. When recovered from Spent, choose which section they enter.

## Setup

For each player:

1. Place 2 Blue stones in Explore.
2. Place 2 Brown stones in Develop.
3. Place 2 Red stones in Settle.
4. Place 2 Green stones in Produce.
5. Place 2 Purple stones in Ship.
6. Place 2 White stones in Spent.
7. Start with 1 Credit.
8. Set up starting Worlds, starting Developments, and queued tiles normally for this prototype.

Each section has a capacity of 6 stones. If no section can receive a stone during sowing, place that stone in Spent.

## Round Structure

Each round:

1. Each player chooses one source: a board section, or one color group in Spent.
2. Each player sows those stones.
3. The final section sown selects that player's phase.
4. All selected phases occur for all players, in normal Roll phase order.
5. Resolve Manage Empire.

There is no Citizenry and no dice assignment area.

## Sowing

When sowing from a board section, pick up every stone in that section and sow clockwise into later sections, one stone per section.

When sowing from Spent, choose one color group and sow any number of those stones. Native colors enter their matching section:

|        |            |
|--------|------------|
| Blue   | -> Explore |
| Brown  | -> Develop |
| Red    | -> Settle  |
| Green  | -> Produce |
| Purple | -> Ship    |

White and Yellow may enter any section.

Skip full sections. If every section is full, overflow stones go to Spent.

## Match Bonuses

If the final stone lands in its matching section, mark one phase bonus for that section this round.

|                  |                                           |
|------------------|-------------------------------------------|
| Blue in Explore  | +1 Explore action                         |
| Brown in Develop | -1 Credit cost this Develop               |
| Red in Settle    | -1 Credit cost this Settle                |
| Green in Produce | +1 Produce action                         |
| Purple in Ship   | +1 VP on the first shipped Good this Ship |

White and Yellow do not create match bonuses.

## Phase Resolution

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Stones stay on the mancala board when phases resolve.

### Explore

Explore actions equal stones in Explore plus the Explore match bonus.

For each action:

1. If your Development queue is empty, scout one Development.
2. Else if your World queue is empty, scout one World.
3. Else gain 2 Credits.

### Develop

Develop completes the top queued Development if you can pay its effective cost.

```
effective cost =
Development cost
- matching Brown stones in Develop
- White/Yellow stones in Develop
- Develop match bonus
```

Pay the remaining cost in Credits. If you cannot pay, nothing happens to that Development this phase. No partial progress is kept.

### Settle

Settle works like Develop, but for the top queued World.

```
effective cost =
World cost
- matching Red stones in Settle
- White/Yellow stones in Settle
- Settle match bonus
```

Pay the remaining cost in Credits. If you cannot pay, nothing happens to that World this phase. No partial progress is kept.

### Produce

Produce actions equal stones in Produce plus the Produce match bonus.

Each action places one Good marker on an eligible empty production World in your tableau. The stone remains in Produce. The Good marker is not a die.

### Ship

Ship actions equal stones in Ship.

Each action ships one Good marker. If you have 2 or fewer Credits, trade the Good for Credits. Otherwise, consume it for VP chips. The Ship match bonus adds +1 VP to the first Good consumed during this Ship phase.

The shipped Good marker is removed. The Ship stone remains in Ship.

## Manage Empire

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During Manage Empire, spend 2 Credits to make one recovery sow from Spent. Repeat while you can pay and still have stones in Spent.

Credits do not become dice. Credits do not enter the mancala board. Credits pay construction costs and recover Spent stones.

## Gaining And Losing Dice

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When a tile grants a die:

1. Put it in its matching section if there is room.
2. Otherwise put it in Spent.

When a tile removes a die:

1. Remove from Spent first, avoiding colors the tile grants if possible.
2. If needed, remove from a board section.

Goods and construction queues are not dice.

## Card Text Conversion

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Use the following translation rules:

- Extra phase worker: add one temporary action or discount for that phase.
- Cost reduction: reduce the Credit cost of the matching Develop or Settle.
- Gain Credits: gain Credits normally, up to the Credit cap.
- Produce a Good: place a Good marker on an eligible World.
- Trade or Consume: use Ship actions and Good markers.
- Reassign a die: move an unused stone between board sections only if the text clearly supports moving a die.
- Dice-count scoring: count physical dice in board sections and Spent only.

Ignore text that would place a die as lasting construction progress. This variant does not track partial builds.

## Multiplayer Notes

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All selected phases are shared. If one player selects Develop, every player may resolve Develop. A player can benefit from a phase even if they did not select it, as long as they have the relevant stones, Credits, queued tile, or Goods.

The current prototype settings are:

|                         |                            |
|-------------------------|----------------------------|
| Section capacity:       | 6                          |
| Starting stones:        | 2 per normal phase section |
| Starting white stones:  | 2 in Spent                 |
| Starting yellow stones: | 0                          |
| Starting Credits:       | 1                          |
| Recovery sow cost:      | 2 Credits                  |

|                      |              |
|----------------------|--------------|
| VP pool:             | 8 per player |
| Tableau end trigger: | 12 cards     |

Recent smoke simulation with four strategies averaged about 10 rounds and showed no recurring dead-round jam.

## Solo Mode

Solo mode uses the same player rules and adds a simple dummy phase deck.

Setup:

1. Build a five-card Dummy Phase deck: Explore, Develop, Settle, Produce, Ship.
2. Shuffle it.
3. Set the VP pool to 30.
4. Play up to 16 rounds.

Each round:

1. You sow and select one phase.
2. Draw 2 Dummy Phase cards.
3. Resolve your selected phase plus the dummy phases, in normal phase order.
4. Resolve the dummy effects.
5. Resolve Manage Empire.
6. If the Dummy deck cannot supply 2 cards, reshuffle all five Dummy Phase cards.

Dummy effects:

|         |                                                               |
|---------|---------------------------------------------------------------|
| Explore | Dummy claims one Development and one World from the tile bag. |
| Develop | Dummy claims one Development.                                 |
| Settle  | Dummy claims one World.                                       |
| Produce | Dummy Goods +1, max 4.                                        |
| Ship    | Drain 2 VP per Dummy Good, minimum 2, then Dummy Goods = 0.   |

Solo ends when any of these happen:

- you reach 12 tableau cards;
- the VP pool is empty;
- round 16 ends.

## Solo Scenarios

Score scenarios:

|            |        |
|------------|--------|
| Great      | 38+ VP |
| Triumphant | 42+ VP |
| Epic       | 46+ VP |

Named scenarios require at least 34 VP plus the listed mark:

|                    |                      |
|--------------------|----------------------|
| Builder            | 7+ completed tiles   |
| Developer          | 4+ Developments      |
| Colonizer          | 6+ Worlds            |
| Shipper            | 8+ VP chips          |
| Expanded Workforce | 17+ owned dice       |
| Producer           | 4+ production Worlds |

|                     |                          |
|---------------------|--------------------------|
| Diversified Economy | 3+ distinct World colors |
| Phase Specialist    | 3+ color match bonuses   |
| Credit Economy      | 18+ Credits spent        |
| Logistics           | 2+ recovery sows         |
| Phase Momentum      | 65+ phase actions        |

Owned dice means physical dice in your mancala sections plus Spent. Good markers and queued tiles do not count. Phase actions are actions produced by stones in resolved phase sections, including bonus actions from match bonuses.

## Solo Campaigns

A campaign is four consecutive games. Each game must mark one campaign scenario not already marked. Win the campaign by marking all four scenarios.

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| Frontier Survey         | Great, Colonizer, Diversified Economy, Phase Specialist  |
| Core Worlds Renaissance | Triumphant, Developer, Builder, Credit Economy           |
| Trade League            | Triumphant, Producer, Shipper, Diversified Economy       |
| Supply Lines            | Great, Expanded Workforce, Logistics, Phase Momentum     |
| Colonial Boom           | Great, Colonizer, Producer, Logistics                    |
| Industrial Mobilization | Epic, Expanded Workforce, Credit Economy, Phase Momentum |
| Imperial Prestige       | Epic, Builder, Shipper, Phase Specialist                 |
| Galactic Mastery        | Epic, Credit Economy, Logistics, Phase Momentum          |

Campaign story hooks:

- Frontier Survey: chart a scattered sector and turn discoveries into a working colony network.
- Core Worlds Renaissance: rebuild the home cluster through Developments, infrastructure, and Credit discipline.
- Trade League: make production and shipping the center of galactic influence.
- Supply Lines: keep a wide empire supplied through workforce growth and repeated recovery.
- Colonial Boom: expand fast, then make the new Worlds produce.
- Industrial Mobilization: push a large workforce and Credit economy into an Epic-scale engine.
- Imperial Prestige: convert visible accomplishments into prestige before the Dummy drains the sector.
- Galactic Mastery: a hard economy-control arc for strong solo play.

Current balanced-strategy tuning target:

|                                                                                     |                                       |
|-------------------------------------------------------------------------------------|---------------------------------------|
| Frontier Survey / Trade League / Colonial Boom:                                     | approachable campaign wins            |
| Core Worlds / Supply Lines / Industrial Mobilization / Imperial Prestige / Mastery: | harder economy and prestige campaigns |

## Current Test Coverage

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The automated tests cover:

- five-section setup and Spent;
- board sowing and Spent recovery;
- phase sharing;
- Develop/Settle Credit costs and section discounts;
- match bonuses;
- Produce Good markers without moving stones;
- Ship Good markers without moving stones;
- solo dummy phase selection;
- solo capacity counting and campaign definitions.